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Gautam Neelakantan Memana

Education

Australian National University 2026-2027
Postdoctoral Researcher Mentor: Po Lam Yung

University of Wisconsin Madison 2021-2026
PhD student
PhD thesis: On nonlinear maximally subelliptic PDEs Advisor: Brian Street

Indian Institute of Science Education and Research, Mohali 2016-2021, CGPA: 9.8/10.0
BS-MS program, Mathematics major
MS thesis: L^p Harmonic analysis on Heisenberg group Advisor: Jotsaroop Kaur

Research

- Gautam Neelakantan Memana, Neil S. Trudinger, Po-Lam Yung "*Fractional elementary symmetric functions and Hessian estimates in the plane*", https://arxiv.org/abs/2610.19410
- Gautam Neelakantan Memana, "*A counterexample to Fourier alignment in single neuron modular addition*", https://arxiv.org/pdf/2608.04451
- Gautam Neelakantan Memana, "*Comparison theorems for weak solutions of nonlinear maximally sub-elliptic PDEs*", https://arxiv.org/abs/2604.12291
- Gautam Neelakantan Memana, "*A Regularity Theorem For Fully nonlinear Maximally Subelliptic PDE*", https://arxiv.org/abs/2409.04344
- Soma Maity, Gautam Neelakantan Memana, "*Uniform Poincaré inequalities on measured metric spaces*", Manuscripta Mathematica (2022)
- Gautam Neelakantan Memana, "L^p *Harmonic analysis on the Heisenberg group*", MS thesis at IISER Mohali

Teaching Experience

- Teaching Assistant:Elementary Matrix and Linear Algebra (MATH 340) Fall 2025
- Teaching Assistant: Introduction to Discrete Mathematics (MATH240) Summer 2025
- Teaching Assistant: Calculus and Analytic Geometry II(Calc II) Fall 2024
- Teaching Assistant:Elementary Matrix and Linear Algebra (MATH 340) Summer 2024
- Instructor: Algebra (MATH 112) Fall 2023
- Instructor: The Theory of Single Variable Calculus(MATH 421) Summer 2023
- Teaching Assistant:Elementary Matrix and Linear Algebra (MATH 340) Spring 2023
- Teaching Assistant: Techniques in Ordinary Differential Equations (MATH 319) Fall 2022
- Direct Reading Program: Ergodic Theory Fall 2022
- Teaching Assistant: Algebra and Trignometry(MATH 114) Spring 2022
- Teaching Assistant: Algebra and Trignometry(MATH 114) Fall 2021

Scholarships and Awards

- AMS sectional travel grant — Fall 2025
- Henry Schaerf Mathematics Graduate Award — Fall 2024
- MinneMUDAC best prediction team award in the graduate section — Spring 2025
- DAAD WISE scholarship — May–July, 2019
- VSRP Scholarship for internship at TIFR CAM, Bangalore — May –July, 2018
- Madhava Mathematics competition invitation to attend camp — Summer 2018
- Kishore Vaigyanik Protsahan Yojana scholarhip — 2017–2021
- INSPIRE Fellowship — 2016–2017

Invited Talks/Prsentations

- NIT Calicut- Mathematics department seminar — August 18, 2026
- Cochin University of Science and Technology- Mathematics department seminar — July 21, 2026
- Workshop OpArt à St Dié des Vosges — May 27, 2026
- AMS special session in Harmonic Analysis- North Dakota State University, Fargo — April 2026
- Contributed talk at AMS Central Sectional Meeting in St. Louis — October 2025
- Cochin University of Science and Technology- Mathematics department seminar — May 2025
- Poster presentation at Madison Lectures in Harmonic Analysis — May 2024
- Kopp summer school- Analysis of multiple ergodic averages — September 2023
- University of Florida-analysis seminar — April 2023
- Kopp summer school- Nodal domains and landscape functions — October 2022
- Poster presentation at Nordfjordeid Summer School 2019 - Analysis, Geometry and PDE — June 2019